

7 (a) Define *magnetic flux*.

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..... [1]

- (b) A rigid conducting loop of diameter 6.0 cm is suspended freely by a non-conducting cord as shown in Fig. 7.1. A uniform magnetic field of flux density 0.20 T is directed at right angles to the plane of the loop. X and X' are two points within the magnetic field.

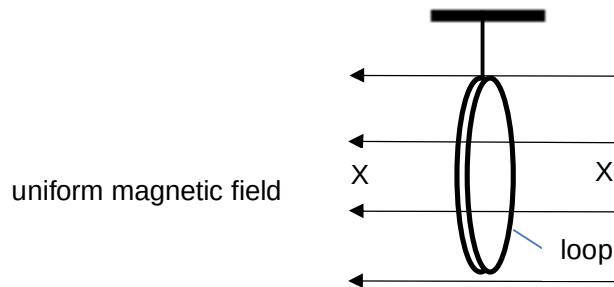


Fig. 7.1

- (i) Determine the magnetic flux through the loop.

magnetic flux = Wb [2]

- (ii) The magnetic flux density is increased to 0.60 T in a time of 0.20 s. An induced current flows in the loop.

1. On Fig. 7.1, indicate the direction of the induced current in the loop. [1]
2. Calculate the average induced current in the loop if its resistance is 0.20 Ω .

induced current = A [2]

- (c) The loop is set to swing like a pendulum within the magnetic field between X and X'.

The loop comes to rest in a short time. Explain why this is so.

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..... [2]

8 With the rising concern in global warming, there is a renewed interest in expanding the